

Title: HebSO

Date:

Master eq.

$$\sum_j A_{ij} = S, \quad s: \text{edge strength} \quad \bar{S}: \frac{S}{n(n-1)}$$

① random pruned: $\frac{1}{n(n-1)}$

② growth: +1

random: $(1-p) \cdot \frac{1}{n(n-1)}$ preferential $p \frac{S}{S}$

$$P_{t+1}(S) - P_t(S) = -\frac{1}{n(n-1)} P_t(S) - \left(\frac{1-p}{n(n-1)} + \frac{pS}{S} \right) P_t(S) \bar{S} \quad (S \text{ 不变})$$

$$+ \left(\frac{1-p}{n(n-1)} + \frac{p(S-1)}{S} \right) P_t(S-1) \bar{S}$$

令 $P_{t+1}(S) = P_t(S)$:

$$\left(\frac{1}{n(n-1)} + \left(\frac{1-p}{n(n-1)} + \frac{pS}{S} \right) \bar{S} \right) P_t(S) = \left(\frac{1-p}{n(n-1)} + \frac{p(S-1)}{S} \right) P_t(S-1) \bar{S}$$

$$\bar{S} / S = 1/n(n-1)$$

$$(1 + (1-p)\bar{S} + pS) \frac{\bar{S}}{S} P_t(S) = ((1-p)\bar{S} + p(S-1)) \frac{\bar{S}}{S} P_t(S-1)$$

$$P_t(S) = \frac{(1-p)\bar{S} + p(S-1)}{1 + (1-p)\bar{S} + pS} P_t(S-1)$$

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$$X = \tanh[\beta(Ax + b)] \quad \beta: \text{inverse temp}$$

$$A_{ij}: \text{coupling} \quad b_i: \text{external field}$$

$$\hat{x} = \sigma[\beta(\tilde{A}\hat{x} + \tilde{b})]$$

$$\textcircled{1} \text{ norm: } \hat{x} = \frac{x+1}{2}$$

$$\textcircled{2} \text{ map}$$

$$\left(\frac{e^{\beta u} - e^{-\beta u}}{e^{\beta u} + e^{-\beta u}} + 1 \right) / 2 = \frac{1}{1 + e^{-2\beta u}}$$

$$\tilde{A}\tilde{x} + \tilde{b} = 2(Ax + b) = 4A\left(\frac{x+1}{2}\right) + b - 2A$$

$$\tilde{x}_i = (x_i + 1)/2 \quad \tilde{A}_{ij} = 4A_{ij} \quad \tilde{b}_i = 2(b_i - \sum_j A_{ij})$$

$$[0,1] \quad [-1,1]$$

$$C_{ij} = \frac{dx_i}{db_j} = \text{sech}[\beta(\sum_k A_{ik}x_k + b_i)]^2 \beta \left(\sum_k A_{ik} \frac{dx_k}{db_j} + \delta_{ij} \right)$$

$$= \beta(1 - x_i^2) \left(\sum_k A_{ik} C_{kj} + \delta_{ij} \right)$$

$$C = \beta D A C + \beta D, \quad C = \beta(I - \beta D A)^{-1} D$$